molecules. Hot tea is tea in which the molecules are very agitated. Cold tea is tea in which the molecules are only a little agitated. In an ice cube, warming up and melting molecules become increasingly agitated and lose their strict connections.

At the end of the nineteenth century, there were many who still did not believe in the existence of molecules and atoms: Ludwig was convinced of their reality and entered the fray on behalf of his belief. His diatribes against those who doubted the reality of atoms became legendary. “Our generation were at heart all on his side,” remarked one of the young lions of quantum mechanics years later.[16] In one of these fiery polemics, at a conference in Vienna, a noted scientist[17] maintained against him that scientific materialism was dead because the laws of matter are not subject to the directionality of time. Scientists are not immune from talking nonsense.

Looking at the sun going down, the eyes of Copernicus had seen the world turning. Looking at a glass of still water, the eyes of Boltzmann saw atoms and molecules frenziedly *moving*.

We see the water in a glass like the astronauts saw the Earth from the moon: calm, gleaming, blue. From the moon, they could see nothing of the exuberant agitation of life on Earth, its plants and animals, desires and despairs. Only a veined blue ball. Within the reflections in a glass of water, there is an analogous tumultuous life, made up of the activities of myriads of molecules—many more than there are living beings on Earth.

This tumult *stirs up* everything. If one section of the molecules is still, it becomes stirred up by the frenzy of neighboring ones that set them in motion, too: the agitation spreads, the molecules bump into and shove each other. In this way, cold things are heated in contact with hot ones: their molecules become jostled by hot ones and pushed into ferment. That is, they heat up.

Thermal agitation is like a continual shuffling of a pack of cards: if the cards are in order, the shuffling disorders them. In this way, heat passes from hot to cold, and not vice versa: by shuffling, by the natural disordering of everything. The growth of entropy is nothing other than the ubiquitous and familiar natural increase of disorder.

This is what Boltzmann understood. The difference between past and future does not lie in the elementary laws of motion; it does not reside in the deep grammar of nature. It is the natural disordering that leads to gradually less particular, less special situations.

It was a brilliant intuition, and a correct one. But does it clarify the difference between past and future? It does not. It just shifts the question. The question now becomes: why, in one of the two directions of time—the one we